

Supplementary Materials

Beyond Fidelity: Evaluating Student Early-Warning Explanations with Proxy Simulatability

These supplementary materials provide implementation details, prompt schematics, educational feature availability, identifier definitions, and localized HELOC rule-edit results.

S1. Black-box and surrogate implementation

The black-box implementation uses fixed, untuned XGBoost configurations. Dropout models use 800 estimators, maximum depth 4, learning rate 0.03, row and column subsampling of 0.85, L_2 regularization 2.0, L_1 regularization 0, minimum child weight 3, `binary:logistic`, log loss, histogram tree construction. HELOC differs only in using a minimum child weight 5. The model seed is 2027. The classifier is fitted only on the black-box training partition, without an evaluation set or early stopping. Consequently, the reserved validation labels are not used for tuning or model selection. We also use the default binary decision rule (probability 0.5).

For the surrogate, a `DecisionTreeClassifier` is fitted to the fixed XGBoost labels on the surrogate partition for each maximum depth in $\{10, 15, 20\}$, with minimum leaf size 15 for dropout and 40 for HELOC and seeds 2037, 2042, and 2047, respectively. Each leaf is traversed depth-first from the left branch ($\leq$ threshold) before the right branch ($>$ threshold), and becomes one rule. Rules are ordered by this traversal; execution sorts by `(priority, rule_id)`, returns the first matching rule, and otherwise returns the modal black-box label on the surrogate partition. Depth is selected on the refinement partition by the verifier objective J . An exact tie retains the first evaluated depth, so the order 10, 15, 20 favors depth 10.

S2. Prompt schematics and verification

The following schematics retain the source’s inputs and output format. Angle-bracketed fields are filled for the relevant dataset and condition. They are not exact run logs. For HELOC, the valid labels are Good and Bad.

S2.1. Rule-refiner prompt

```

1  SYSTEM
2  You are a careful rule-refinement assistant. You edit symbolic
3  rules only through the allowed JSON DSL. Do not invent features
4  or labels. Return JSON only. Do not include hidden reasoning,
5  markdown, or <think> text.
6
7  USER
8  Refine one weak rule from a surrogate rule set for a fixed
9  black-box classifier.
10 Counterexample source: <faithfulness | proxy>
11 Feature schema: <valid labels, feature types, descriptions,
12                 allowed values or training ranges>
13 Current full rule set: <ordered rules and default label>
14 Weak rule to edit: <rule identifier, conditions, conclusion>
15 Weak rule statistics: <support, precision, predicate count,
16                      and proxy disagreement count if used>
17 Examples: <up to eight relevant records with black-box label,
18           rule label, source, and first firing rule; proxy
19           examples also include proxy label, confidence,
20           and short reason when available>
21
22 Allowed operations: add_condition, remove_condition,
23 change_threshold, split_rule, merge_rules, change_default,
24 or abstain.
25
26 Return a JSON array with at most three candidate operations.
27 Every condition must contain exactly feature, op, and value.
28 Do not include prose outside JSON.
```

S2.2. Main-text verification schematic

```

1  current <- R0
2  for iteration in 1..8:
3    weak_rules <- rank at most 6 rules using
4                  the selected source
5    accepted <- empty
6
```

```

7   for rule in weak_rules:
8       examples <- select at most 8 records
9       proposals <- first 3 JSON edits returned
10          by the refiner
11
12   for edit in proposals:
13       candidate <- apply edit to a copy of current
14       reject malformed edits, invalid labels,
15          features or operators, out-of-range
16          thresholds, duplicate identifiers,
17          impossible bounds, or complexity-
18          budget violations
19       execute candidate on refinement partition
20
21       if coverage(candidate) >= 0.90 and
22          conflict_rate(candidate) <= 0.05 and
23          rule_count(candidate) <= 64 and
24          mean_predicates(candidate) <= 8 and
25          total_predicates(candidate) <= 320 and
26          J(candidate) - J(current) > 0.0005:
27           add candidate to accepted
28
29   if accepted is empty:
30       break
31
32   current <- candidate in accepted with highest J
33   # At most one edit is adopted per iteration.
34
35   return current

```

S2.3. Proxy-simulator prompt

```

1  SYSTEM
2  You are simulating a fixed black-box classifier from the
3  provided input and optional explanation. Predict the black-box
4  label, not the ground-truth label. Return JSON only.
5
6  USER
7  {
8      "valid_labels": ["Dropout", "Non Dropout"],
9      "condition": "<x-only | R0 | refinement | local method>",
10     "mode": "<forward | counterfactual>",
11     "feature_schema": <readable schema>,
12     "x": <source student record>,
13     "explanation": <rules, local explanation, or
14                   "No rule explanation is provided.">,
15     "original_black_box_label": <counterfactual mode only>,
16     "x_counterfactual": <counterfactual mode only>,
17     "required_output_format": {
18         "predicted_black_box_label": "one valid label",
19         "confidence": "number from 0.0 to 1.0",
20         "rule_ids_used": ["rule identifiers if any"],
21         "short_reason": "one sentence only"
22     }
23 }

```

S3. Educational dataset feature inventory and identifier meanings

The main paper describes fourteen educational datasets, Data_1–Data_14, abbreviated here as D1–D14. Table 9 reports the source feature identifiers and their availability in each dataset. A checkmark records field presence. Table 10 provides the corresponding identifier meanings from the experiment’s feature dictionary. Identifying, target, and other administrative fields remain listed so that the source inventory can be audited against the actual modeling schema.

Table 9. Source field availability across the fourteen educational datasets (D1–D14). Checkmarks indicate that a field is present, not that it was used as a model input.

Feature	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
Adress	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
AdresseL	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
CD_REG	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Classment_class	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_1000	✓	✓	✓	✓	✓	✓								
Coefficients.CC_1033											✓			
Coefficients.CC_111											✓		✓	
Coefficients.CC_11	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_12	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_131													✓	
Coefficients.CC_132													✓	
Coefficients.CC_133													✓	
Coefficients.CC_134													✓	
Coefficients.CC_13										✓	✓	✓	✓	✓
Coefficients.CC_14											✓		✓	
Coefficients.CC_15											✓		✓	
Coefficients.CC_16											✓		✓	
Coefficients.CC_18						✓	✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_19	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_20							✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_23							✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_24	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_26							✓	✓	✓	✓	✓	✓	✓	✓
Coefficients.CC_30										✓	✓			
Coefficients.CC_34	✓	✓	✓	✓	✓	✓								
Coefficients.CC_36										✓	✓	✓	✓	✓
Coefficients.CC_37												✓	✓	
Coefficients.CC_60										✓	✓	✓	✓	✓
Coefficients.CC_95													✓	
DO_ETAB	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Exist_Internet	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
INDHcom	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
INDHquart	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Internat	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Level	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MCaRtable	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MoyenneClasse	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MoyenneGen	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Multiclass	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Nb_class_dropout	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Nb_class_failure	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NbrJourAbsenceAutorise	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NbrJourAbsenceNonAutorise	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NbrUniteAbsenceAutorise	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NbrUniteAbsenceNonAutorise	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_1000	✓	✓	✓	✓	✓	✓								
NoteCC_1033											✓			
NoteCC_111											✓		✓	
NoteCC_11	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_12	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_131													✓	
NoteCC_132													✓	
NoteCC_133													✓	
NoteCC_134													✓	
NoteCC_13										✓	✓	✓	✓	✓
NoteCC_14											✓			

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Feature	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
NoteCC_15											✓		✓	
NoteCC_16											✓		✓	
NoteCC_18						✓	✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_19	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_20							✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_23							✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_24	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_26							✓	✓	✓	✓	✓	✓	✓	✓
NoteCC_30										✓	✓			
NoteCC_34	✓	✓	✓	✓	✓	✓								
NoteCC_36										✓	✓	✓	✓	✓
NoteCC_37												✓	✓	
NoteCC_60										✓		✓	✓	
NoteCC_95													✓	
NoteExam_11						✓			✓					
NoteExam_12						✓			✓					
NoteExam_18						✓			✓					
NoteExam_19						✓			✓					
NoteExam_20									✓					
NoteExam_23									✓					
NoteExam_24						✓			✓					
NoteExam_34						✓								
ProgrammeTissir	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Préscolarisé	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
White_year	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
absence_exam_11						✓			✓					
absence_exam_12						✓			✓					
absence_exam_18						✓			✓					
absence_exam_19						✓			✓					
absence_exam_20									✓					
absence_exam_23									✓					
absence_exam_24						✓			✓					
absence_exam_34						✓								
cd_etab	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
cd_prov	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
datenaiseleve	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
failure	✓	✓	✓	✓	✓	✓	✓	✓	✓					
id_annee	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
id_classe	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
id_eleve	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
id_genre	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
id_handicap	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
international							✓	✓	✓	✓		✓	✓	✓
istayssir	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
nationalite	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
nbr_eleves	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
nbr_filles	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
profession_mere	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
profession_pere	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
target	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
typeEtab	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 10. Meanings of the source feature identifiers

Identifier	Meaning
Adress	Student address
AdresseL	School address
CD.REG	Region name
Classment.class	Student's class ranking
Coefficients.CC.1000	Coefficient for Artistic Education
Coefficients.CC.1033	Coefficient for vocational subjects
Coefficients.CC.111	Coefficient for Artistic Culture
Coefficients.CC.11	Coefficient for Arabic Language
Coefficients.CC.12	Coefficient for French Language
Coefficients.CC.131	Coefficient for Accounting and Financial Mathematics
Coefficients.CC.132	Coefficient for General Economics and Statistics
Coefficients.CC.133	Coefficient for Economics and Business Administration
Coefficients.CC.134	Coefficient for Management Information Systems
Coefficients.CC.13	Coefficient for English Language
Coefficients.CC.14	Coefficient for Spanish Language
Coefficients.CC.15	Coefficient for German Language
Coefficients.CC.16	Coefficient for Italian Language
Coefficients.CC.18	Coefficient for History and Geography
Coefficients.CC.19	Coefficient for Mathematics
Coefficients.CC.20	Coefficient for Life and Earth Sciences
Coefficients.CC.23	Coefficient for Physics and Chemistry
Coefficients.CC.24	Coefficient for Islamic Studies
Coefficients.CC.26	Coefficient for Physical Education
Coefficients.CC.30	Coefficient for Computer Science
Coefficients.CC.34	Coefficient for Scientific Activities
Coefficients.CC.36	Coefficient for Philosophy
Coefficients.CC.37	Coefficient for Translation
Coefficients.CC.60	Coefficient for Attendance and Conduct
Coefficients.CC.95	Coefficient for Law
DO.ETAB	School opening date
Exist.Internet	Whether the student had internet access
INDHcom	Whether the commune is in a precarious situation
INDHquart	Whether the neighbourhood is in a precarious situation
Internat	Boarding-school status
Level	Current educational level
MCarTable	Participation in the Cartable school-supplies programme
MoyenneClasse	Mean grade of the class
MoyenneGen	Average final grade
Multiclass	Whether multiple educational levels are taught in the same class
Nb.class.dropout	Number of dropouts in the class
Nb.class.failure	Number of failures in the class
NbrJourAbsenceAutorise	Number of authorised absence days
NbrJourAbsenceNonAutorise	Number of unauthorised absence days
NbrUniteAbsenceAutorise	Number of authorised missed classes
NbrUniteAbsenceNonAutorise	Number of unauthorised missed classes
NoteCC.1000	Final average grade for Artistic Education
NoteCC.1033	Final average grade for vocational subjects
NoteCC.111	Final average grade for Artistic Culture
NoteCC.11	Final average grade for Arabic Language
NoteCC.12	Final average grade for French Language
NoteCC.131	Final average grade for Accounting and Financial Mathematics
NoteCC.132	Final average grade for General Economics and Statistics
NoteCC.133	Final average grade for Economics and Business Administration
NoteCC.134	Final average grade for Management Information Systems
NoteCC.13	Final average grade for English Language
NoteCC.14	Final average grade for Spanish Language
NoteCC.15	Final average grade for German Language
NoteCC.16	Final average grade for Italian Language
NoteCC.18	Final average grade for History and Geography
NoteCC.19	Final average grade for Mathematics
NoteCC.20	Final average grade for Life and Earth Sciences
NoteCC.23	Final average grade for Physics and Chemistry
NoteCC.24	Final average grade for Islamic Studies
NoteCC.26	Final average grade for Physical Education
NoteCC.30	Final average grade for Computer Science

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Table 10 – continued

Identifier	Meaning
NoteCC_34	Final average grade for Scientific Activities
NoteCC_36	Final average grade for Philosophy
NoteCC_37	Final average grade for Translation
NoteCC_60	Final average grade for Attendance and Conduct
NoteCC_95	Final average grade for Law
NoteExam_11	Arabic Language exam grade
NoteExam_12	French Language exam grade
NoteExam_18	History and Geography exam grade
NoteExam_19	Mathematics exam grade
NoteExam_20	Life and Earth Sciences exam grade
NoteExam_23	Physics and Chemistry exam grade
NoteExam_24	Islamic Studies exam grade
NoteExam_34	Scientific Activities exam grade
ProgrammeTissir	Participation in the Tissir financial-aid programme
Préscolarisé	Whether the student attended preschool
White-year	Gap year
absence_exam_11	Number of final exams missed for Arabic Language
absence_exam_12	Number of final exams missed for French Language
absence_exam_18	Number of final exams missed for History and Geography
absence_exam_19	Number of final exams missed for Mathematics
absence_exam_20	Number of final exams missed for Life and Earth Sciences
absence_exam_23	Number of final exams missed for Physics and Chemistry
absence_exam_24	Number of final exams missed for Islamic Studies
absence_exam_34	Number of final exams missed for Scientific Activities
cd.etab	School identifier
cd.prov	Province name
datenaيسةleve	Date of birth
failure	Whether the student failed the previous year
id.annee	School year
id.classe	Class identifier
id.eleve	Student identifier
id.genre	Sex
id.handicap	Disability category
international	International Baccalaureate status
istayssir	Participation in Tayssir, a financial-support programme intended to keep students in school
nationalite	Nationality
nbr.eleves	Number of students in the class
nbr.filles	Number of female students in the class
profession.mere	Mother's occupation
profession.pere	Father's occupation
target	Dropout status
typeEtab	School type

S3. Reported HELOC rule edits

The following table preserves all edits and refinement/evaluation counts reported in the source. Affected decisions and affected firing paths are different quantities. The same-label path changes do not imply altered predictions.

Table 11. Localized HELOC rule edits relative to R0. Only modified rules are shown. Behavioral effects report the number of affected records on the refinement/evaluation partitions ($n = 1,046$ each). We abbreviate the features `AverageMinFile` as A , `NumSatisfactoryTrades` as N , `PercentTradesWBalance` as P , and `PercentTradesWBalance_is_special` as S_P ; $S_P = 1$ denotes a special or missing value and $S_P = 0$ an ordinary observed value.

Refinement	Rule(s)	Edit relative to R0	Behavioral effect (refinement/evaluation)
Faithful-Refined	r_007	Add $P \leq 80$ and $S_P = 0$.	11/18 predictions change from Good to the default Bad.
Faithful-Refined	r_009	Replace $A > 63.5$ with $A > 65$.	1/0 predictions change from Good to the default Bad.
Faithful-Refined	r_010	Add $P \leq 85$.	1/7 predictions change from Good to the default Bad.
Proxy-Refined	r_009 {r_009_s0, r_009_s1}	Replace $A > 63.5$ with $A > 65$, then split the rule into $N \leq 10.5$ and $N > 10.5$ branches.	1/0 predictions change; 279/311 records formerly firing r_010–r_012 now fire r_009_s1, with the Good conclusion unchanged.